

WEB SCRAPING PROJECT

AmbitionBox Reviews Data Extraction & Analysis

A Python-based project for extracting, processing, and analyzing company review data from the AmbitionBox website.

Python • BeautifulSoup • Requests • Pandas

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Process of Web Scraping

Web Scraping Process

Web scraping is the automated process of extracting data from websites. In this project, data is collected from the AmbitionBox Reviews website using Python libraries such as Requests, BeautifulSoup, and Pandas.

Step 1: Website Selection

The AmbitionBox Reviews website was selected as the target source for extracting company review information and related data.

Step 2: Sending HTTP Request

A request is sent to the website using the Requests library to retrieve the HTML content of the web page.

Step 3: Parsing HTML Content

The downloaded HTML page is parsed using BeautifulSoup, which allows easy navigation and extraction of required elements from the webpage.

Step 4: Inspecting Web Elements

The HTML structure of the webpage is inspected to identify relevant tags, classes, and attributes containing the required information.

Step 5: Extracting Data

Relevant information such as company names, ratings, reviews, and other available details are extracted from the identified HTML elements.

Step 6: Data Cleaning

The extracted data is cleaned by removing unnecessary characters, duplicate records, and missing values to improve data quality.

Step 7: Creating DataFrame

The cleaned data is organized into a structured Pandas DataFrame for easy analysis and manipulation.

Step 8: Exporting Data

The final dataset is exported and stored in CSV format for future use and analysis.

Step 9: Data Analysis

The collected data can be analyzed using statistical methods and visualized using charts and graphs to gain meaningful insights.

Outcome

The web scraping process successfully demonstrates how data can be collected from websites, processed, organized, and prepared for further analysis using Python.

Step 1 : Import Required Libraries

```
In [ ]: import requests
from bs4 import BeautifulSoup
import pandas as pd
```

Step 2 : Define Target Website URL

```
In [2]: url = "https://www.ambitionbox.com/reviews"

headers = {
    "User-Agent": "Mozilla/5.0"
}
```

Step 3 : Send HTTP Request

```
In [3]: response = requests.get(url, headers=headers)

print("Status Code:", response.status_code)
```

Status Code: 200

Step 4 : Parse HTML Content using BeautifulSoup

```
In [4]: soup = BeautifulSoup(response.text, "html.parser")

print(soup.title)

<title>Company Reviews and Ratings By Employees | AmbitionBox</title>
```

Step 6 : Extract Required Information

```
classes = set()

for tag in soup.find_all(): if tag.get("class"): classes.update(tag.get("class"))

print("Total Classes Found:", len(classes))

for cls in sorted(classes): print(cls)
```

Step 7 : Store Extracted Data into Python Lists

```
In [6]: reviews = []

for item in soup.find_all("h2"):
    text = item.get_text(strip=True)

    if text:
        reviews.append(text)

print("Total Records:", len(reviews))
```

Total Records: 5

Step 8 : Create Pandas DataFrame

```
In [7]: df = pd.DataFrame(
    reviews,
    columns=["Review Data"]
)

df.head()
```

```
Out[7]:
```

	Review Data
0	Browse Companies by Location
1	Company Reviews
2	Browse Companies by Industries
3	Browse by Company Size
4	Popular Companies

Step 9 : Export Data to CSV File

```
In [8]: df.to_csv(
    "ambitionbox_reviews.csv",
    index=False,
    encoding="utf-8"
)

print("CSV File Saved Successfully")
```

CSV File Saved Successfully

Results

- Successfully connected to AmbitionBox website.
- Parsed HTML using BeautifulSoup.
- Extracted available review information.
- Stored data in CSV format.

Conclusion

This project demonstrates the use of Python, Requests, BeautifulSoup, and Pandas for web scraping and data storage. The extracted data can be used for further analysis and visualization.